

Finite Wing Aerodynamics

- All real wings are finite in length.
- Elementary flow models have been only applied to an ideal wing (i.e. Airfoil) also known as an Infinite wing.
- Here we introduce the physical effects that come with a finite wing span.
- For example, the **NACA 4412** airfoil at the **8-degree** angle of attack will have a lift coefficient of **1.2** and a drag coefficient of **0.0068**.
- But the 3D wing with NACA 4412 airfoil will not have the same lift and drag coefficients (**because of the 3D flow effect**).
- We know that

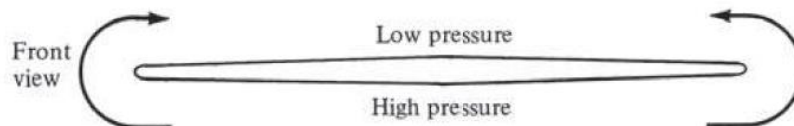
- $C'_l = \frac{L'}{(1/2)\rho V^2 c}$ for an airfoil $S = c \cdot 1$

- $C_L = \frac{L}{(1/2)\rho V^2 S}$ for a wing

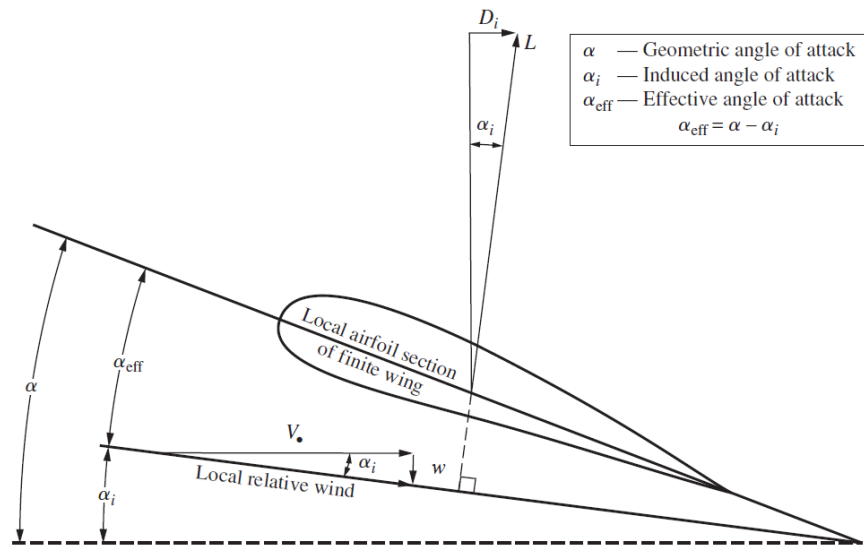
- “Per unit span” is an ideal case, assuming $S = \infty$
- In real cases, $S \neq \infty$, leads to finite-span effects.

Downwash and Induced drag:

- Flow leaks from bottom to top cause tip vortices.



- Tip vortices induce downwash (vertical velocity or induced velocity).
- Vertical velocity components due to tip vortices are called induced velocity.
- When we add the vertical velocity w with the freestream velocity component, the resultant total velocity vector has an angle α_i with respect to the travel direction.
- Effective angle of attack can be written as, $\alpha_{\text{eff}} = \alpha - \alpha_i$
- Lift force generated is now perpendicular to the new total velocity vector U_{eff} and no longer perpendicular to U_{∞} .
- Change in the relative wind reduce the vertical lift and adds a new drag in the direction of motion called induced drag.
- Induced drag – due to downwash tilted the lift vector.



Effects of downwash:

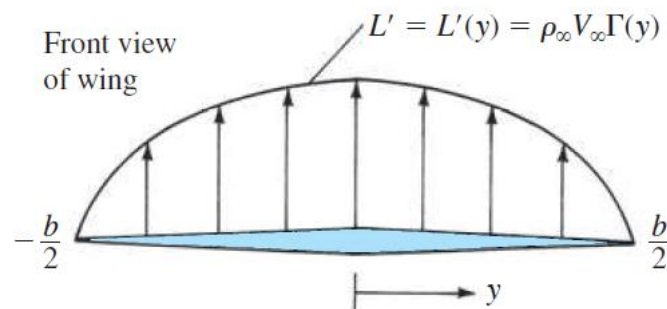
1. Change in the angle of attack locally (**decrease in lift**).
2. Portion of lift in the direction of motion becomes induced drag (**Addition of drag**).

For a finite wing, there are three main sources of drag:

1. Skin-friction drag (**due to viscous effect**)
2. Pressure or Form drag (**due to pressure imbalance**)
3. Induced drag (**due to downwash**)

} **Profile drag**

- Downwash leads to lift distribution.



- From Kutta-Joukowski Theorem, Lift per unit span is varying across the span, which means the circulation also varies.
- Downwash is not the only physical phenomenon that causes lift distribution.

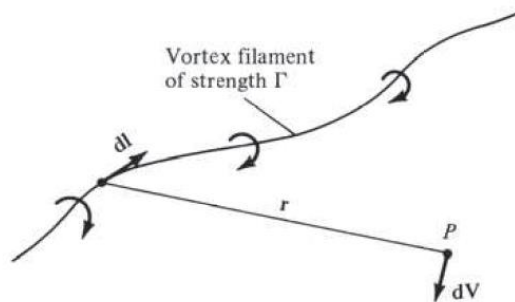
There are other factors that cause lift distribution.

- Chord variation (**Taper ratio = tip chord/ root chord**)
- Geometric twist (**change in twist angle across the span**)
 - Wash out (**wing tip at a lower angle than the root**)

- Wash in (**wing root at a lower angle than the tip**)
- Aerodynamic twist (**different value of zero-lift angle of attack $\alpha_{L=0}$ across the span**). **Root foil is different from tip foil.**

Biot – Savart Law:

- Most fundamental relation in the theory of inviscid, incompressible flow.
- Biot – Savart law is a general result of **Potential theory**.
- Can be used to understand the effect of semi-infinite vortices.



Potential Theory:

- Study of harmonic functions (solutions of Laplace equation).

Potential flow:

- In fluid dynamics, potential flow is described by means of velocity potential (ϕ), being a function of space and time.

$$\vec{V} = \nabla \phi$$

- All the elementary flows and their combinations are potential flows.
- They satisfy the Laplace equation
- They are irrotational, incompressible or compressible, and Inviscid.

From Biot – Savart Law, for a vortex filament with strength Γ the induced velocity due to segment ds at an arbitrary point P ,

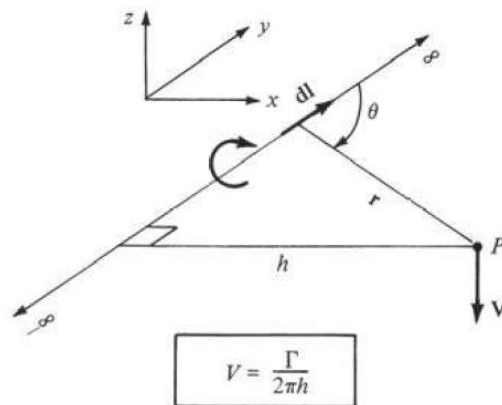
$$d\vec{V} = \frac{\Gamma}{4\pi} \frac{d\vec{l} \times \vec{r}}{|\vec{r}|^3}$$

The induced velocity at point P due to the straight vortex filament from $-\infty$ to $+\infty$

$$\vec{V} = \int_{-\infty}^{+\infty} \frac{\Gamma}{4\pi} \frac{d\vec{l} \times \vec{r}}{|\vec{r}|^3}$$

We need to make some geometric changes to the equation,

h – Perpendicular to the vortex filament.



We know that,

$$\vec{A} \times \vec{B} = |\vec{A}| |\vec{B}| \sin \theta$$

We just need the magnitude of the induced velocity,

$$V = |\vec{V}|$$

Therefore, the equation for the induced velocity becomes,

$$V = \frac{\Gamma}{4\pi} \int_{-\infty}^{\infty} \frac{\sin \theta}{r^2} dl$$

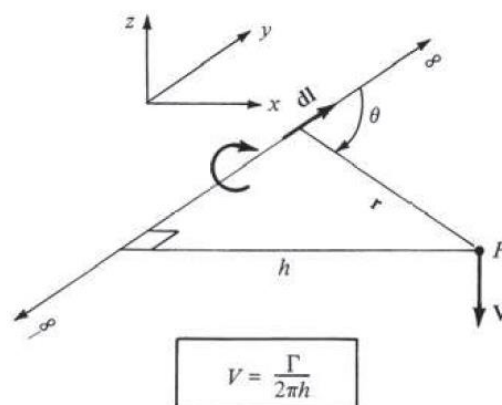
Changing from r and l to h and θ

$$r = \frac{h}{\sin \theta}$$

$$l = \frac{h}{\tan \theta}$$

$$dl = \frac{-h}{\sin^2 \theta} d\theta$$

$$-\infty \text{ to } +\infty \rightarrow \pi \text{ to } 0$$



Therefore,

$$V = -\frac{\Gamma}{4\pi} \int_{\pi}^0 \sin \theta d\theta$$

Equation for the induced velocity for an infinite vortex becomes,

$$V = \frac{\Gamma}{2\pi h}$$

Vertical velocity due to infinite vortex at a distance (perpendicular to filament) h .

- Induced velocity linearly increases with the vortex strength, Γ .
- Induced velocity linearly decreases with the perpendicular distance (h).

Tip vortex:

- Tip vortex has a starting point (i.e. finite)
- Tip vortex does not extend from $-\infty$ to ∞
- Tip vortex extends from 0 to ∞ so we call it as **semi-infinite vortex**.

Therefore,

$$V = \frac{-\Gamma}{4\pi} \int_{\pi}^{\pi/2} \sin \theta d\theta$$

$$V = \frac{\Gamma}{4\pi} [\cos \theta]_{\pi}^{\pi/2}$$

For a semi-infinite vortex,

$$V = \frac{\Gamma}{4\pi h}$$

This equation will be integrated into the calculation to add the induced velocity and downwash effect.

There are rules to vortex filament:

1. Vortex filament strength is constant along the filament (interpreted as the vortex filament strength is not changing in time without external forcing).
2. A vortex filament cannot end in a fluid, it must extend to the boundaries of the fluid (**e.g. tip vortices on the wing**) or it can connect itself and form a closed loop (**Vortex ring**).
3. If nothing causes rotation externally, then irrotational flow stays irrotational.

These rules are theorems, and they are called **Helmholtz Theorems**. They have been shown to be true through measurements and observations.